Module 02 (Assignment)

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Quality Control and Genome Assembly

Quality assessment of the human breast cancer (Stage IIIA) whole genome sequencing dataset (SRR33505761) was conducted using FastQC (Galaxy Version 0.74+galaxy1) on the Galaxy platform (v24.1.4). The dataset comprised 1,156,013 Illumina paired-end reads, totaling 53.1 Mbp of sequence data with an average read length of 46 bp and a GC content of 48%. The FastQC module reported no sequences flagged as poor quality and no adapter contamination, indicating a clean and high-quality dataset. The encoding type was confirmed as Sanger/Illumina 1.9, ensuring compatibility with downstream analytical tools. These metrics confirm that the sequencing data were of excellent quality, with uniform base composition and sufficient depth for subsequent genome assembly.

Following quality assessment, adapters were removed using Cutadapt (v5.1+galaxy0) with a Q20 cutoff and a minimum read length of 50 bp to ensure retention of high-confidence reads. The trimmed reads were then assembled using the SPAdes assembler (Galaxy Version 4.2.0+galaxy0) in assembly and error-correction mode with default parameters (Antipov et al., 2016). Assembly evaluation with QUAST (Galaxy Version 5.3.0+galaxy1) generated key metrics including N50, L50, number of contigs, largest contig, total assembly size, and GC content, confirming that the data were of sufficient quality for accurate de novo assembly. Completeness of the assembly was further validated using BUSCO (Galaxy Version 5.8.0+galaxy2).

Assembly and QC Summary

Metric	Value
Total Reads	1,156,013
Total Bases	53.1 Mbp
Average Read Length	46 bp
%GC Content	48%
Encoding	Sanger / Illumina 1.9
Sequences Flagged as Poor Quality	0
Adapter Contamination	None detected
Assembler Used	SPAdes (v4.2.0+galaxy0)
# Contigs (>= 0 bp)	2,854
# Contigs (>= 500 bp)	48
Largest Contig	8,606 bp
Total Length (>= 0 bp)	207,522 bp
N50	1,231 bp
L50	11
GC %	50.02%
Ns per 100 kbp	0

Interpretation: The high-quality FastQC metrics and clean adapter profile indicate that the input reads were suitable for de novo genome assembly. SPAdes successfully generated contigs, though the N50 (1,231 bp) and the largest contig (8,606 bp) suggest that the assembly was fragmented, likely due to the short read length (46 bp). The GC content (50.02%) falls within the expected range for human genomic data. Zero N bases confirm that the assembly contains no undefined regions. The overall data are appropriate for downstream analyses such as variant calling, functional annotation, and structural assessment.

References

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